

E-LEARNING PLATFORM WITH ADAPTIVE QUIZ GENERATION USING ITEM RESPONSE THEORY

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ABSTRACT

This study develops an adaptive e-learning platform that uses Item Response Theory (IRT) to personalize quiz difficulty based on estimated student ability, improving learning outcomes by 18% over fixed-difficulty quizzes.

INTRODUCTION

Standardized quizzes in e-learning systems fail to adapt to individual student ability levels, leading to frustration for struggling learners and boredom for advanced ones. Item Response Theory (IRT) provides a psychometric framework for dynamic difficulty adjustment. This study implements a 2-parameter IRT model in a web-based e-learning platform for BSIT programming courses at CvSU.

METHODOLOGY

A bank of 300 multiple-choice questions for 5 programming topics was calibrated using 2PL IRT (discrimination and difficulty parameters estimated via maximum likelihood). The adaptive algorithm uses Expected A Posteriori (EAP) ability estimation and selects the next item to maximize Fisher information. The system was built in Django with a Vue.js frontend. A quasi-experimental study compared adaptive (n=45) vs. fixed-quiz (n=45) groups over one semester. Pre-test and post-test scores measured learning gain.

RESULTS

The adaptive group showed 18.3% higher normalized learning gain than the fixed-quiz group ($p < 0.01$, Cohen's $d = 0.72$). Average quiz completion time was 12.4 minutes (adaptive) vs. 17.8 minutes (fixed), a 30% reduction. Student satisfaction was 4.3/5 (adaptive) vs. 3.7/5 (fixed). Item exposure rate was more balanced in adaptive testing (reduced overexposure of easy items by 41%).

DISCUSSION

IRT-based adaptive testing meaningfully improves learning efficiency and satisfaction compared to fixed quizzes. The statistically significant effect size ($d = 0.72$) indicates practical pedagogical value. The primary implementation challenge was expert calibration of initial IRT parameters. Future work should automate parameter estimation via online EM as more

student responses accumulate.